RecruiterBrain

Intelligent Candidate Discovery & Ranking Engine

Participant: SRIKRISHNA S (krishn9706@gmail.com) Challenge: India Runs x Hack2Skill — Data & AI Challenge

The Paradigm Shift: Recruiter-First Design

Why Traditional Systems Fail:

• Keyword Stuffing: Weak candidates list 30 skills on their resume
without having ever worked on them.

• Silent Experts: Top-tier talent describe their projects naturally
and skip listing dozens of acronym buzzwords.

• Disqualification Traps: Challenge rules automatically disqualify entries if >10% of the top 100 are faked 'honeypot' profiles.

The RecruiterBrain Solution:

• 5-Layer Pipeline: We combine career trajectories, verified experience bands, skill authenticity weightings, and engagement signals.

• Deterministic Reasoning: Every rank decision is accompanied
by a unique, factual justification explaining why they were chosen.

• Pure Stdlib Speed: Built on standard Python, executing end-to-end on 100K candidates in < 2 minutes with zero external
network or GPU needs.

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System Architecture: 5-Layer Pipeline

Our architecture mirrors the step-by-step screening methodology of an expert recruiter:

Layer Scoring Focus Weight Aesthetic & Technical Objective

L1 + L2: Career Fit

Title relevance, stability (avg tenure > 1.5 yrs), Tier-1 target cities, and product vs
consulting indicators.

40%

Filters out non-technical roles and consultants. Targets 5-9 years experience band preferred in JD.

L3: Skills Intelligence

Matches against Must-Have & Nice-to-Have skills, with duration and endorsement
scaling.

30%

Detects keyword-stuffers by checking if skills are backed by actual career history durations.

L4: Behavioral Signals

Recruiter response rate, active days, notice period (sub-30 preferred), and GitHub
presence.

20%

Prioritizes active, reachable candidates who are highly likely to accept an offer.

L5: Honeypot Filter

Catches profiles with impossible work dates or expert skills with zero duration.

Hard
Filter

Crushes fakes down to ~0.01 score,
guaranteeing a 0% honeypot rate in the top 100.

Bonus: Education

Tier-1 institutions (IITs, NITs, BITS, etc.) and Computer Science related degrees.

10%

Adds a clean differentiator for candidates with
solid foundations.

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Deep Dive: Career Fit (40%) & Skills (30%)

Career Trajectory Analysis

• Title Relevance Matrix: Evaluates historical titles (e.g.,
AI/ML/NLP engineers mapped to 1.0, data analysts to 0.5,
marketing to 0.0).

• Tenure Stability: Deducts score for candidates averaging < 1.5
years per job to filter out job-hoppers.

• Production Evidence: Scans career history descriptions for key phrases showing they built production systems (e.g., 'deployed to production', 'RAG pipeline', 'latency', 'vector db').

Advanced Skills Intelligence

• Weight Scale: Must-Have skills (embeddings, vector search, Python) scored 1.0; Nice-to-Have (MLflow, PyTorch) scored 0.6.

• Trust Multiplier: Multiplies skill score by duration in months and
endorsements to reward deep, verified experience.

• Anti-Stuffing Penalty: Penalizes candidates listing expert skills
that never appear in their work history descriptions. **Caught
5,132 stuffing profiles**.

Anti-Keyword Stuffing Rule: Skill score = (Claimed Skills matched in description) / (Total claimed expert skills). If the candidate lacks evidence, their claimed expert status is discounted.

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Behavioral Signals, Honeypots & Reasoning

Behavioral & Availability Signals (20%)

• Notice Period: Prefers immediate joiners or sub-30 day notices
(1.0). Deducts score smoothly for 60/90/120 day notices (down to
0.4).

• Engagement: High multiplier for recruiter response rates > 70%
and recent platform log-in dates.

• GitHub Activity: Rewards candidates who link an active Github repository.

L5: Hard Honeypot Filter

• Contradiction Checker: Detects profiles claiming 10+ years experience in libraries that only existed for 3 years (e.g.,
LangChain expert since 2012).

• Time Travelers: Catches profiles with overlapping timelines (e.g.,
working full-time at two distinct companies in different locations
simultaneously).

• Effect: Instantly multiplies composite score by 0.01, ensuring 0%
honeypots make the final cutoff.

Judges Focus - Plagiarism & Hallucination: Every candidate in our top-100 contains a 100% custom reasoning summary outlining their exact current role, specific experience metrics, and notice period limitations.

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Results, Speed & Compliance Metrics

Evaluation Metric Observed Result Challenge Constraint / Target

Candidates Processed 100,000 candidates Full Dataset

Total Processing Time 114.86 seconds < 5.0 minutes (300 seconds) on 16GB CPU

Honeypots in Top 100 0.0% (0 caught in top 100, 8 total) < 10.0% Disqualification Threshold

Keyword Stuffers caught 5,132 profiles penalized Not specified (Judges filter criteria)

Output File Validity PASSED validate_submission.py Strict CSV format, 100 rows, candidate IDs match

Compute Dependencies

Pure Python Standard Library (0 external APIs, no GPU)

16GB RAM, offline (no network access during run)

Top Candidates Quality: Our top-ranked candidates represent Senior AI/NLP/Applied ML Engineers at top-tier product tech companies (Zomato, Ola, Google, Meta, CRED) with an average of 5-8 years of experience, highly aligned with the JD.

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Interactive Sandbox & Production Visual Dashboard

We have deployed our solution on two environments to allow judges to review and test the ranking engine live:

1. Hugging Face Spaces Sandbox (Required)

URL: https://huggingface.co/spaces/SRI-KRISHNA7/RecruiterBrain

• Interactive Weight Customizer: Real-time slider controls to
dynamically recalculate scores.
• Sample File Processing: Pre-loaded with a subset of 50
candidates, allowing you to test runs on-the-fly and inspect candidate highlights.
• CSV Exporter: Generate and export submission-ready CSVs instantly from the UI.

2. Production Visual Dashboard (Hosted on Vercel)

URL: https://recruiter-brain.vercel.app

• High-Fidelity UI: Premium glassmorphism dark-theme dashboard visualizing the top 100 results.
• Analytics Panel: Provides data charts displaying experience
spans, score counts, and title frequencies.
• Candidate Inspector Modal: Deep resume summaries, skill
badges, experience timelines, and AI justification text.

Submission Links ready for evaluation: CSV output, YAML metadata, public GitHub repo, and both live environments are complete.

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